Tariff-Related Executive Social Media Posts and Next-Day US Equity and Implied Volatility Reactions

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[1]Hypothesis × Quant

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Abstract

Hypothesis: H_0: next-day S&P 500 absolute abnormal return, signed abnormal return and VIX change after a tariff-related post equal those of ordinary days; H_1: they differ
Data: 428 Truth Social posts matching the keyword 'tariff', 237 distinct reaction days among 1,117 sessions (2022-05-12 to 2026-10-23)
Method: Reaction window is the first session strictly after the post; baseline is the trailing 250-day mean; 5000-draw placebo over non-event days; Bonferroni $\alpha = 0.0167$
Results: abs return: event 0.736% vs other 0.724% (placebo $p = 0.772$); signed return: event -0.023% vs other 0.026% (placebo $p = 0.375$); vix change: event 0.810% vs other 0.073% (placebo $p = 0.070$). Decision: **REJECTED**

1 Design

Posts are selected by keyword only, so no text classifier can leak sentiment. Using the session after the post date guarantees that the measured reaction is not contaminated by the same-day move that may have prompted the post [MacKinlay, 1997]. Event days are not independent when posts cluster; the placebo draws random days from the same sample rather than assuming normal errors.

2 Results

3 Limitations

Next-day windows discard the intraday reaction. Keyword matching ignores context. Truth Social coverage starts in 2022, so tariff episodes before that are missing.

Table 1: Reaction on the first trading day after a tariff post

Measure	Event days	Other days	Placebo p
|AR| S&P 500 (%)	0.736	0.724	0.7720
AR S&P 500 (%)	-0.023	0.026	0.3747
ΔVIX (%)	0.810	0.073	0.0696

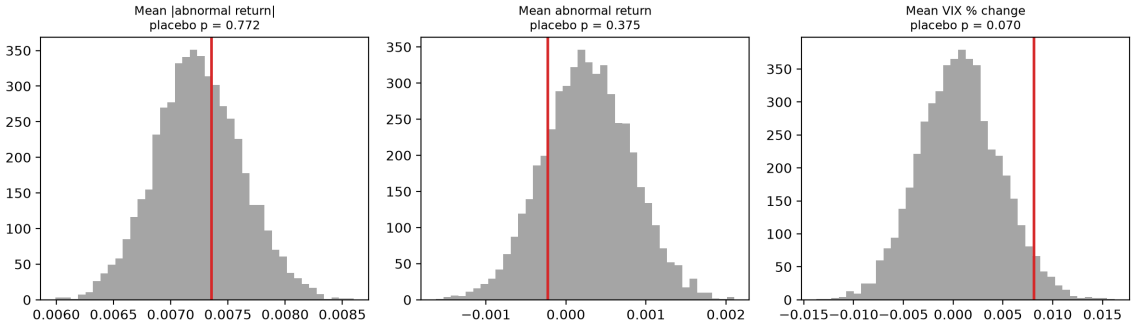


Figure 1: Observed event-day mean (red) against the placebo distribution.

References

A Craig MacKinlay. Event studies in economics and finance. *Journal of Economic Literature*, 35 (1):13–39, 1997.